

# Hesam Ossanloo

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Principal Engineer & Engineering Leader building AI-powered platforms, scalable ecosystems, and enterprise-grade distributed systems.

## WORK EXPERIENCE (Last 14 Years)

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### Autodesk

Oslo, Norway

*Engineering Manager/Principal Engineer*

Feb 2025 – present

- Leadership & Organizational Impact:
  - Lead and mentor 6 engineers (Junior to Principal), driving technical excellence, career growth, and performance calibration.
  - Rebuilt and scaled the technical hiring process, increasing hiring throughput from 1 hire every two months to 2 hires per month while raising technical quality standards.
  - Introduced structured feedback and performance frameworks (Core Quadrant, Multipliers by Liza Wiseman), contributing to a 30% increase in employee satisfaction.
  - Elevated engineering culture by strengthening technical assessment rigor and cross-team calibration practices.
- Technical Leadership & Architecture:
  - Tech Lead for Autodesk Forma Ecosystem: APIs, MCPs, Extensions, Agentic AI integrations.
  - Architect enterprise-grade authentication and authorization systems (OAuth 2.0), ensuring scalability, reliability, and regulatory compliance.
  - Initiated and led the MCP proof of concept for Autodesk University (AU), now in production, which resulted in the formation of a dedicated MCP squad under my leadership.
  - Drive architectural strategy across cloud-native, serverless microservices and micro-frontend systems following the CAP theorem.
  - Established operational excellence practices (Datadog, PagerDuty, AI-driven support tooling), improving reliability and incident response maturity.

### SkogApp/TAVANA

Oslo, Norway

*Cofounder and Chief Product & Technology Officer*

Jan 2023 – Feb 2025

- Defined product vision and technical strategy for two AI-driven ventures in forestry and healthcare.
- Built and led cross-functional teams to develop scalable web/mobile platforms and production ML systems.
- **SkogApp**: Reduced forestry plan creation time from two years to 30 minutes through satellite-image ML models and geospatial analytics.
- **TAVANA**: Built Norway's first medical-domain LLM, reducing documentation time by 30–60 minutes per doctor per day, in collaboration with Harvard Medical School and MIT.

### Antler

Oslo, Norway

*Entrepreneur in Residence*

Feb 2023 – May 2023

- Participated in a rigorous three-month startup program provided by Antler, focused on building and scaling early-stage ventures, and successfully secured pre-seed investment.
- Won the Google Hackathon award for building the most secure and scalable technical solution in the program.

**Spark Sport NZ**  
*Head of Engineering*

**Auckland, New Zealand**  
Feb 2022 – Jun 2023

- Led and mentored a 15-person engineering organization, defining architecture strategy and delivery standards for a multi-platform sports streaming ecosystem serving up to 1M users.
- Implemented CI/CD and automated testing, enhancing product quality and reducing deployment time and cloud-computing costs by 30%.
- Fostered a collaborative team environment, improving team satisfaction and retention rates.

**Mercury NZ Ltd.**  
*Solution Architect / Tech Lead (Contractor)*

**Auckland, New Zealand**  
Jul 2021 – Feb 2022

- Led the modernization of a legacy system by designing a microservice architecture and implementing scalable, secure RESTful APIs using serverless technologies, resulting in a 40% cost reduction.
- Saved the company a potential \$2 million loss, by figuring out the compatibility issue of the new energy platform (Bidgely) and the current tech stack.

**Centrality.ai Ltd. / P^werFinance**  
*Solution Architect / Tech Lead (Contractor)*

**Auckland, New Zealand**  
Aug 2020 – Aug 2021

- Led a team of 3 developers while remaining hands-on in architecture design and code reviews, overseeing DevOps collaboration and production delivery using TDD, BDD, CI/CD, and Agile methodologies.
- Interacted with stakeholders, product owners, and designers to ensure business requirements were accurately mapped to technical assets and skill sets, ensuring efficient, secure, and automated product delivery.
- Architected and implemented a blockchain-based banking platform, including secure minting and burning of encrypted digital assets.

**Propellerhead Ltd.**  
*Senior Fullstack Developer*

**Auckland, New Zealand**  
Oct 2018 – Aug 2020

- Mentored junior developers and collaborated with stakeholders and designers to develop and maintain cloud-based mission control software for Auckland Transport, ensuring real-time data flow and fleet management.
- Reduced public transport messaging/rerouting process by 50%, by implementing a Kafka- and Redis-based live GPS tracking solution with retry mechanisms, to automatically send messages to desired stations.
- Developed and maintained internal applications, including a badge framework for tracking career progression, a spending application, and a revenue forecasting tool.

**INFOMOTION GmbH**  
*Senior IT Consultant (Contractor)*

**Frankfurt, Germany**  
Jun 2018 – Oct 2018

- Automated data extraction from SAP reports, by implementing the “Easy Cockpit” portal, which led to a more user-friendly data presentation for managers and saved time.

**OBJECT ECM GmbH**  
*Senior IT Consultant*

**Zurich, Switzerland**  
Apr 2016 – Jun 2018

- Led a team of junior and intermediate developers, managing various projects and ensuring alignment with stakeholder requirements.
- Reduced construction application processing time by 50%, by implementing an internal web portal and a workflow engine for the Thüringer Aufbaubank, which led to 50% higher state-based support for the citizens.

**Sopra Steria GmbH***Fullstack Developer***Hamburg, Germany**

Feb 2014 – Apr 2016

- Reduced credit fraud by 40%, by implementing a web-based fraud prevention tool and rule-engine for banks and investment companies, in a team of 3 developers.

**NTT DATA Deutschland***Web Developer***Munich, Germany**

Nov 2012 – Feb 2014

- Optimized after-sales communication for BMW and Audi through implementation of integrated web applications, resulting in a 30% reduction in waiting times and operating costs.

**PROJECTS**

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**Project SkogApp (Individual project):**

- Development of a responsive web application (React and TypeScript) where foresters can log in via OAuth 2.0. Based on their roles, users can use a six-step wizard to find and analyze their forest on the map.
- Design of a microservice architecture where AWS Lambda functions (Python and Node.js) download high-resolution satellite or aerial images of the desired property via REST APIs. A pre-trained ML model analyzes the image and creates a forestry plan and a bio-growth model of the forest.
- Personal responsibility for the microservice architecture, product design and REST APIs in the backend as well as for using PostgreSQL/PostGIS and Firestore as databases.

**GitHub Source Code:**

- <https://github.com/hesamossanloo/skogapp-api>
- <https://github.com/hesamossanloo/skogapp-frontend>

**Project Spark Sport (Team project):**

- Built and managed a team of 7 developers and 8 testers to build a live sports streaming app for 16 different platforms (Web, iOS, Android, 13 different smart TVs and streaming devices like Chromecast, Fire TV).
- Used React and TypeScript for the frontend and Node.js and TypeScript for the backend.
- Designed microservice architecture, REST APIs, integration with Snowflake, PagerDuty, DataDog and AI-based anomaly detection for monitoring and logging.

**VOLUNTEER EXPERIENCE**

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**Planetary Transportation Systems GmbH***Data Scientist/Senior Developer***Berlin, Germany**

Sep 2017 – Feb 2020

- Semantic search research for the 'Mission to the Moon' project, an initiative from the Google Lunar X-Prize to put a lunar rover on the moon.
- Accomplished secure data transmission, by building mission control software, which led to secure communication between the Audi lunar Quattro, the Autonomous Landing and Navigation Module (ALINA), and the Earth.

## EDUCATION

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### Doctoral Research, Informatics

*Hochschule Darmstadt – University of Applied Sciences*

**Darmstadt, Germany**

Jul 2016 – Dec 2022

- Focus: Semantic Web, Ontology and Taxonomy, Machine learning, AI, ETL, Search result clustering, Linked Open Data, Information Retrieval, Semantic Information Visualization.

### M.Sc. of Computer Science (Dual Master)

*Hochschule Darmstadt / Wisconsin Platteville USA*

**Darmstadt, Germany**

Oct 2010 – Oct 2012

- Focus: Enterprise Architecture, Fullstack development (Java, JS, Ruby), Business Process Optimization.

## SKILLS AND TECHNOLOGIES

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- **AI & Intelligent Systems:** LLM systems, MCP architectures, Agentic AI integrations, Semantic Search, Ontology Engineering, Data Privacy & AI Governance.
- **Architecture & Distributed Systems:** Enterprise Architecture, Cloud-Native Systems, Serverless Microservices, Micro-Frontends, Event-Driven Architecture, RESTful APIs, CAP Theorem Applications.
- **Security & Identity:** OAuth 2.0, Authentication & Authorization Architecture, Enterprise Compliance & Secure API Design.
- **Cloud & DevOps:** AWS, Azure, GCP, Kubernetes, Docker, CI/CD, Infrastructure as Code, Observability (Datadog, PagerDuty).
- **Backend & Data:** Node.js, TypeScript, Python, Java, PostgreSQL/PostGIS, DynamoDB, Kafka, Redis, Snowflake.
- **Leadership:** Engineering Management, Technical Strategy, Organizational Scaling, Hiring & Calibration Systems, Mentorship, Cross-Functional Alignment.

## PUBLICATIONS

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- Bernhard G. Humm, Hesam Ossanloo: Domain-Specific Semantic Search Applications: Example SoftwareFinder. In: Thomas Hoppe, Bernhard G. Humm, Anatol Reibold (Eds.): Semantic Applications - Methodology, Technology, Corporate Use. pp. 243-258. Springer Verlag, Berlin, 2018. ISBN 978-3-662-55432-6.
- Bernhard G. Humm, Hesam Ossanloo: Cost-Effective Semi-Automatic Ontology Development from Large Domain Terminology. In: Udo Bleimann, Bernhard Humm, Robert Loew, Stefanie Regier, Ingo Stengel, Paul Walsh (Eds): Proceedings of the Collaborative European Research Conference (CERC 2017), pp 73-82, Karlsruhe, Germany, 22-23 September 2017. ISSN: 2220-4164.
- Bernhard G. Humm and Hesam Ossanloo (2016). A Semantic Search Engine for Software Components. In: Pedro Isaías (Ed.): Proceedings of the International Conference WWW/Internet 2016, IADIS Press, pp 127-135, Mannheim, Germany, ISBN 978-989-8533-57-9.

## LANGUAGES AND HOBBIES

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- Languages: English (Fluent), German (Fluent), Persian (Native), Norwegian (Basics), Arabic (Basics), Turkish (Basics).

- Hobbies: Solving LeetCode puzzles, Reading scientific literature, Mentoring, Running (I used to be a professional 800m Track & Field Runner), Football, Badminton, Ping-Pong